

Bharat Badgujar

Linux Module 2

Assignment no 4

1.

Check the current SELinux status on your Linux server using the `getenforce` and `sestatus` commands, and write down which mode (Enforcing, Permissive, or Disabled) is currently active.

Ans : **Check the current SELinux status**

Run the following commands:

`getenforce`

`sestatus`

Example Output

Enforcing

or

Permissive

or

Disabled

Example `sestatus` output:

SELinux status: enabled

Current mode: Enforcing

Mode from config file: Enforcing

Policy version: 33

Policy from config file: targeted

Current Mode: Enforcing

2.

Switch SELinux to Enforcing mode using `setenforce` and verify the change with `getenforce`.

***Hint:** You may need sudo privileges to change the mode.*

Ans : **Switch SELinux to Enforcing mode**

Command:

```
sudo setenforce 1
```

Verify:

```
getenforce
```

Example Output:

```
Enforcing
```

3.

Simulate an SELinux access denial by creating a new directory (e.g., /tmp/testapp), setting its context to a type not allowed for your web server (such as user_home_t), and then try to serve a file from it using your web server. Note the error message in /var/log/audit/audit.log or /var/log/messages.

Ans : **Simulate an SELinux access denial**

Create a test directory

```
sudo mkdir -p /tmp/testapp
```

Create a sample web page

```
echo "SELinux Test Page" | sudo tee /tmp/testapp/index.html
```

Assign an incorrect SELinux context

```
sudo chcon -t user_home_t /tmp/testapp
```

Verify:

```
ls -Zd /tmp/testapp
```

Example:

```
system_u:object_r:user_home_t:s0 /tmp/testapp
```

Restart Apache:

```
sudo systemctl restart httpd
```

or (Ubuntu/Debian)

```
sudo systemctl restart apache2
```

Try opening the page in a browser.

Example:

```
http://localhost/testapp/index.html
```

The web server should return:

403 Forbidden

Check the SELinux log

```
sudo tail -20 /var/log/audit/audit.log
```

Example log:

```
type=AVC msg=audit(1718000000.123:455): avc: denied { read } for pid=1543
```

```
comm="httpd"
```

```
name="index.html"
```

```
scontext=system_u:system_r:httpd_t:s0
```

```
tcontext=system_u:object_r:user_home_t:s0
```

```
tclass=file
```

4.

Use the `chcon` command to assign the correct SELinux context (e.g., `httpd_sys_content_t`) to your test directory or file so your web server can access it, then confirm the issue is resolved by reloading the page successfully.

Ans : **Fix the SELinux context**

Assign the proper context:

```
sudo chcon -R -t httpd_sys_content_t /tmp/testapp
```

Verify:

```
ls -Zd /tmp/testapp
```

Example:

```
system_u:object_r:httpd_sys_content_t:s0 /tmp/testapp
```

Restart the web server:

```
sudo systemctl restart httpd
```

or

```
sudo systemctl restart apache2
```

Reload the webpage.

Expected result:

SELinux Test Page

The page should now load successfully.

5.

Document the steps you took to identify and resolve the SELinux access denial, including the commands used, the error log entry, and a brief explanation of how SELinux contexts control access.

Ans : **Commands Used**

```
getenforce
sestatus
sudo setenforce 1
sudo mkdir -p /tmp/testapp
echo "SELinux Test Page" | sudo tee /tmp/testapp/index.html
sudo chcon -t user_home_t /tmp/testapp
ls -Zd /tmp/testapp
sudo tail -20 /var/log/audit/audit.log
sudo chcon -R -t httpd_sys_content_t /tmp/testapp
sudo systemctl restart httpd
```

Error Log Entry

```
type=AVC msg=audit(1718000000.123:455): avc: denied { read } for pid=1543
comm="httpd"
name="index.html"
scontext=system_u:system_r:httpd_t:s0
tcontext=system_u:object_r:user_home_t:s0
tclass=file
```